LECTURE: SEPARATION OF VARIABLES (I)

Today: Welcome to the most important PDE technique of the course: Separation of Variables. It's *the* technique that will help us solve lots of PDE, like Laplace's equation.

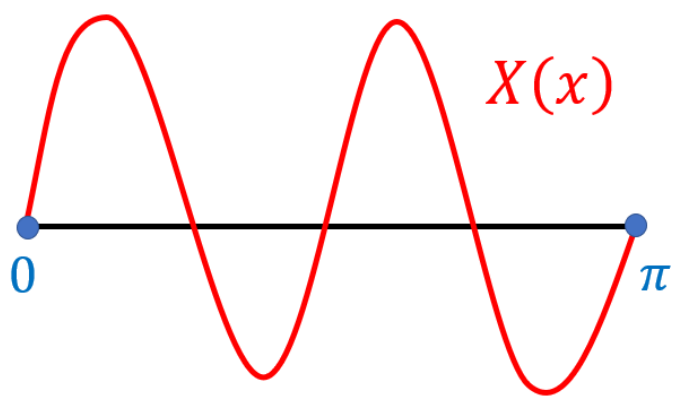
1. Boundary-Value Problem

Before we do that, let's solve a problem that is purely ODE but that will be *crucial* in our process

Example 1:

Find the values of λ for which the ODE has a **nonzero** solution

$$\begin{cases} X''(x) = \lambda X(x) \\ X(0) = 0 \\ X(\pi) = 0 \end{cases}$$



Auxiliary Equation: $r^2 = \lambda$

Case 1: $\lambda > 0$

Notice that if $\lambda > 0$ then $\lambda = \omega^2$ for some $\omega > 0$ (frequency)

Ex: If $\lambda = 9 = 3^2$ then $\omega = 3$

Aux: $r^2 = \lambda = \omega^2 \Rightarrow r = \pm\omega$ in which case we get

$$\begin{aligned} X(x) =& Ae^{\omega x} + Be^{-\omega x} \\ X(0) =& Ae^0 + Be^0 = A + B = 0 \Rightarrow B = -A \\ X(x) =& Ae^{\omega x} - Ae^{-\omega x} \end{aligned}$$

$$\begin{aligned} X(\pi) =& 0 \\ Ae^{\omega\pi} - Ae^{-\omega\pi} =& 0 \\ \cancel{A}e^{\omega\pi} =& \cancel{A}e^{-\omega\pi} \quad A \neq 0 \text{ because otherwise } X(x) = 0 \\ \omega\pi =& -\omega\pi \\ 2\pi\omega =& 0 \\ \omega =& 0 \end{aligned}$$

But then $\lambda = \omega^2 = 0^2 = 0$, which contradicts $\lambda > 0 \Rightarrow\Leftarrow$

Conclusion: In this case, we have no nonzero solutions

Case 2: $\lambda = 0$

Aux: $r^2 = 0 \Rightarrow r = 0$ (repeated twice)

$$\begin{aligned} X(x) =& Ae^{0x} + Bxe^{0x} = A + Bx \\ X(0) =& A + B(0) = A = 0 \\ \\ X(x) =& Bx \\ X(\pi) =& 0 \Rightarrow B\pi = 0 \Rightarrow B = 0 \end{aligned}$$

But then $X(x) = 0x = 0 \Rightarrow\Leftarrow$ (since we want nonzero solutions)

Conclusion: In this case, we also have no nonzero solutions

Case 3: $\lambda < 0$

In this case $\lambda = -\omega^2$ where $\omega > 0$

Ex: $\lambda = -9 = -3^2$ so $\omega = 3$

Aux: $r^2 = \lambda = -\omega^2 \Rightarrow r = \pm\omega i$

$$\begin{aligned} X(x) =& A\cos(\omega x) + B\sin(\omega x) \\ X(0) =& A\cos(0) + B\sin(0) = A = 0 \end{aligned}$$

$$\boxed{X(x) = B\sin(\omega x)}$$

$$\begin{aligned} X(\pi) =& 0 \\ \cancel{B}\sin(\omega\pi) =& 0 \qquad B \neq 0 \text{ because otherwise } X(x) = 0\sin(\omega x) = 0 \\ \sin(\omega\pi) =& 0 \\ \omega\pi =& \pi m \\ \omega =& m \qquad m \text{ is any integer} \end{aligned}$$

Since $\omega > 0$, m needs to be any **positive** integer: $m = 1, 2, 3, \cdots$

Then $\lambda = -\omega^2$

Moreover, from the above, we get $X(x) = B\sin(\omega x) = B\sin(mx)$

Conclusion:

Eigenvalues: $\lambda = -m^2$ $(m = 1, 2, \cdots)$

Eigenfunctions: $X(x) = \sin(mx)$ $(m = 1, 2, \cdots)$

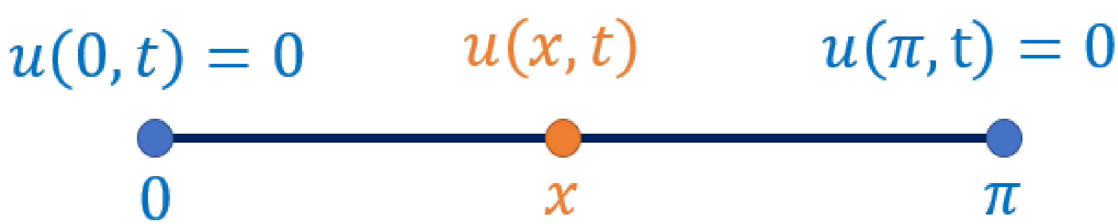
Note: We ignore the constant B here because we'll reintroduce it later

2. Separation of Variables

Video: Separation of Variables

Example 2:

$$\begin{cases} u_t = Du_{xx} & \\ u(0,t) = 0 & \text{(Endpoint)} \\ u(\pi,t) = 0 & \text{(Endpoint)} \\ u(x,0) = x^2 & \text{(Initial)} \end{cases}$$



It's the finite rod problem that is insulated at the endpoints, but this time we want to solve it.

Goal: We want to find **nonzero** solutions

PART I: SEPARATION OF VARIABLES

STEP 1: Assume $u(x,t)$ has a special form, namely

$$\boxed{u(x,t) = X(x)T(t)}$$

Where $X(x)$ is a function of x and $T(t)$ is a function of t

Ex: $\sin(x)\cos(t)$ is of this form, but $\ln(3x+2t)$ is not

Note: There's no reason why u *has* to have that form, but this assumption will be useful to turn the PDE into ODEs, see below

STEP 2: Plug this into the PDE

$$\begin{aligned} u_t &= D u_{xx} \\ \left(X(x)T(t)\right)_t &= D\left(X(x)T(t)\right)_{xx} \\ X(x)\left(T(t)\right)_t &= D\left(X(x)\right)_{xx} T(t) \\ X(x)\,T'(t) &= D X''(x) T(t) \end{aligned}$$

STEP 3: Put the X terms on one side and the T terms on the other

$$\frac{T'(t)}{DT(t)} = \frac{X''(x)}{X(x)}$$

Note: Always put any extra terms like D or c^2 with T, because we want to keep the X equation as simple as possible

PART II: CONSTANT

STEP 1: Look at the quantity

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{DT(t)}$$

$$\text{On the one hand } \left(\frac{X''(x)}{X(x)}\right)_t = 0 \quad \text{(only depends on } x\text{)}$$

$$\text{On the other hand } \left(\frac{X''(x)}{X(x)}\right)_x = \left(\frac{T'(t)}{DT(t)}\right)_x = 0$$

$$\text{Therefore } \frac{X''(x)}{X(x)} \text{ doesn't depend on } x \text{ or } t$$

$$\text{Hence } \frac{X''(x)}{X(x)} = \frac{T'(t)}{DT(t)} = \text{ Constant } = \lambda$$

This implies

$$\frac{X''(x)}{X(x)} = \lambda \Rightarrow X''(x) = \lambda X(x)$$

$$\frac{T'(t)}{DT(t)} = \lambda \Rightarrow T'(t) = D\lambda T(t)$$

This is good news! Instead of a PDE (hard), we get two ODEs (easier)

STEP 2: Focus on the X equation

Note: Don't touch T until the end!!!

So far: $X''(x) = \lambda X(x)$

Now use the boundary conditions:

$$\begin{aligned} u(0,t) =& 0 \\ X(0)\cancel{T(t)} =& 0 \qquad T(t) \neq 0 \text{ because otherwise } u = XT = 0 \\ X(0) =& 0 \end{aligned}$$

$$\text{Similarly } u(\pi,t) = 0 \Rightarrow X(\pi)\cancel{T(t)} = 0 \Rightarrow X(\pi) = 0$$

Hence we get the ODE

$$\begin{cases} X''(x) =& \lambda X(x) \\ X(0) =& 0 \\ X(\pi) =& 0 \end{cases}$$

Which is the same ODE as before!! **WOW!!**

PART III: BOUNDARY-VALUE PROBLEM

Insert Previous Example

$$\boxed{\begin{cases} \lambda = -m^2 & (m = 1, 2, \cdots) \\ X(x) = \sin(mx) & (m = 1, 2, \cdots) \end{cases}}$$

PART IV: T EQUATION

$$\begin{aligned} \frac{T'(t)}{DT(t)} =& \lambda = -m^2 \\ T'(t) =& -m^2 DT(t) \\ T(t) =& e^{-m^2 Dt} \end{aligned}$$

Note: Ignore the constant C for now, it will appear later

Hence for every $m = 1, 2, \cdots$ the following is a solution of our PDE

$$\boxed{u(x,t) = X(x)T(t) = \sin(mx)e^{-m^2Dt}}$$

PART V: LINEARITY

Important Observation: Since the PDE is **linear**, any linear combo of the above solution is also a solution!

$$u(x,t) = A_1 \sin(1x)e^{-(1)^2Dt} + A_2 \sin(2x)e^{-(2)^2Dt} + \cdots$$

$$\boxed{u(x,t) = \sum_{m=1}^{\infty} A_m \sin(mx)e^{-m^2Dt} \text{ is a solution}}$$

PART VI: INITIAL CONDITION

$$u(x,0) = \sum_{m=1}^{\infty} A_m \sin(mx)e^{-m^2D0} = \sum_{m=1}^{\infty} A_m \sin(mx)$$

$$x^2 = \sum_{m=1}^{\infty} A_m \sin(mx)$$

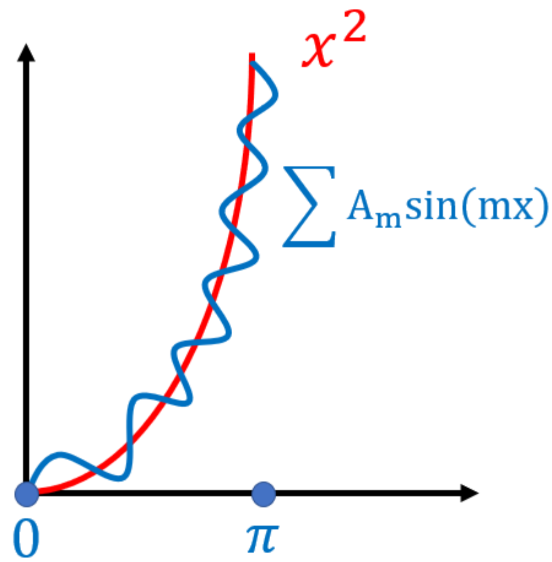
Fourier Question:

Can you find A_m such that

$$x^2 = \sum_{m=1}^{\infty} A_m \sin(mx)$$

In other words, can you write x^2 as a linear combo of sines?

Answer: **YES!!!** This will be the point of some of the later lectures



3. Easier Problem

Example 3:

$$\begin{cases} u_t = Du_{xx} \\ u(0,t) = 0 \\ u(\pi,t) = 0 \\ u(x,0) = 3\sin(2x) + 4\sin(3x) \end{cases}$$

Important Remark: If your initial condition is already in terms of $\sin(mx)$ then you don't need Fourier series!

The previous steps still work and we still get

$$u(x,t) = \sum_{m=1}^{\infty} A_m \sin(mx) e^{-m^2 Dt}$$

$$\text{This time } u(x,0) = \sum_{m=1}^{\infty} A_m \sin(mx) = 3\sin(2x) + 4\sin(3x)$$

Using the definition of a sum we get

$$\begin{array}{lllll} & A_1 \sin(x)+ & A_2 \sin(2x)+ & A_3 \sin(3x)+ & A_4 \sin(4x)+ & \cdots \\ = & 0 \sin(x)+ & 3 \sin(2x)+ & 4 \sin(3x)+ & 0 \sin(4x)+ & \cdots \end{array}$$

Comparing coefficients we get

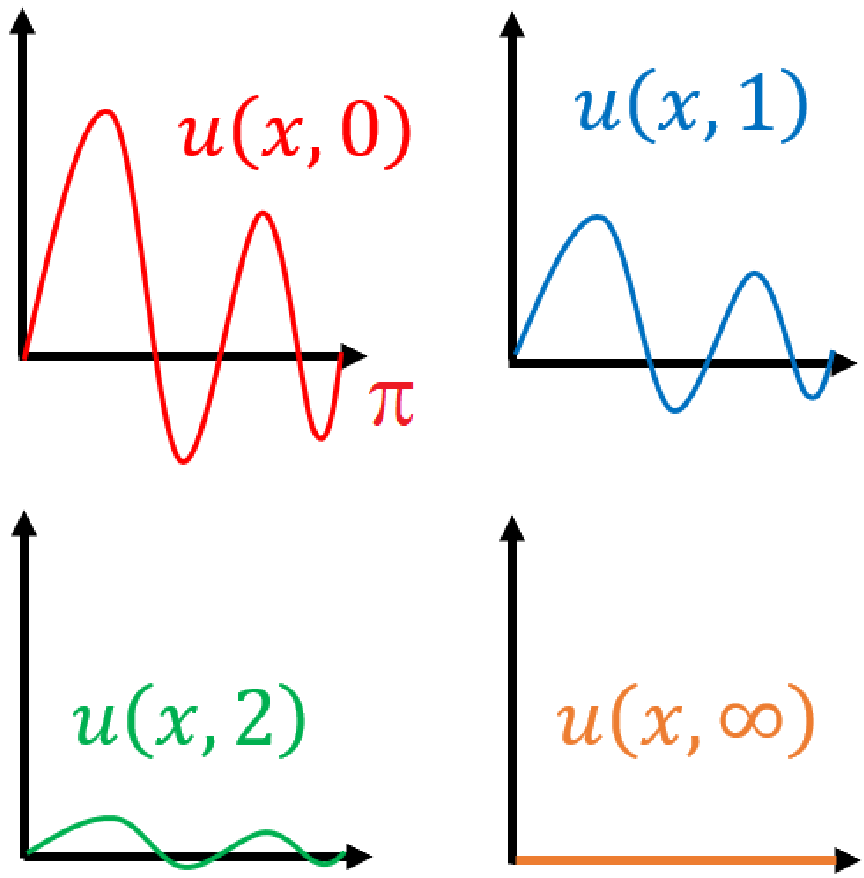
$$A_1 = 0 \text{ and } \quad A_2 = 3 \text{ and } \quad A_3 = 4 \text{ and } \quad A_4 = 0 \cdots$$

And therefore

$$u(x,t) = \sum_{m=1}^{\infty} A_m \sin(mx) e^{-m^2 Dt} = 3\sin(2x) e^{-4Dt} + 4\sin(3x) e^{-9Dt}$$

Interpretation:

Initially, at $t = 0$, the solution starts out as $3\sin(2x) + 4\sin(3x)$ but eventually dies down due to the exponential terms going to 0



Demo: Separation of Variables Demo